

Overview.

This file serves as the bridge between the raw literature and the bibliometric network maps presented in the manuscript. By manually extracting, normalizing, and clustering keywords from all 110 primary studies published between 2021 and 2025, this quantifies the conceptual evolution of machine learning-based insider threat detection. It organizes the academic vocabulary into distinct thematic clusters—such as Core Problem Domains, Algorithmic Architectures, and Data Handling—providing the structured frequency data required to reproducibly generate visualizations that illustrate the field's shift toward complex deep learning architectures and advanced preprocessing techniques.

#	Author (Year)	Key Words
1	Adun (2023)	Threat, Machine Learning, Training, Support Vector Machine
2	Ahmad (2025)	Anomaly Detection; Data Exfiltration; UBA; SIEM; Machine Learning; Insider Threats
3	Ahmadi (2025)	Zero Trust Architecture (ZTA) Segmentation Behavioral Analytics Dynamic Access Governance Machine Learning Insider Threat Detection
4	Alabdulkreem (2022)	Ubiquitous computing Sustainable environment Privacy Data leakage detection Deep learning Ensemble classifier
5	Ali (2025)	insider threat; natural language processing; NLP; classification; machine learning.
6	Al-Mhiqani (2021)	Security; insider threat; insider threats detection; machine learning; deep learning; imbalanced data
7	Al-Mhiqani (2022)	Insider threat; Cybersecurity; Classification; Entropy; VIKOR
8	Almusawy (2024)	machine learning, classification algorithms, synthetic minority oversampling technique, Community Emergency Response Team (CERT) dataset
9	Al-Shehari T and Alsawail (2021) One-hot Endcoding	data leakage detection; one-hot encoding; oversampling technique; insider threat detection; machine learning model
10	Al-shehri Abdullah (2022) Relational Deep Learning	Insider threats; machine learning; recurrent neural network; user behavior analytic
11	Al-Shehari T et.al (2023) Isolation Forest	Anomaly detection, dataset imbalance issue, insider threat detection, isolation forest, machine learning
12	Al-Shehari and Alsawail (2023) Random resampling	Insider data leakage detection ,Imbalanced data classification ,Resampling techniques , Machine learning model
13	Al-Shehari et al (2024) comparative evaluation + CNN	Deep learning, CNN, SMOTE, ADASYN, Data imbalance addressing, Insider threat detection
14	Al-Shehari et al (2024) LOF	Machine learning, insider threat detection, local outlier factor algorithm, data imbalance addressing
15	AlSlaiman (2023)	Insider threat detection, Deep learning, CERT dataset R4.2, Long short term memory LSTM, Sentiment analysis, Gray encoding
16	Amiri-Zarandi (2023)	Insider threat detection, Internet of things, Trustworthy learning, Privacy, Security, Federated learning, Explainable artificial intelligence, Interpretable artificial intelligence, Anomaly detection
17	Amuda (2022)	Insider threats, Deep learning, Convolutional Neural Network, Gated Recurrent Unit, User Behaviour

18	Anakath (2022)	Cloud computing; security; insider attack; network security; privacy; user interaction behavior; deep belief neural network
19	Anju (2024)	Modified equilibrium optimization; Bi-directional gated recurrent unit; Stacked convolutional neural network; Insider threat; User data
20	Asha (2024)	Anomaly detection; Cloud security; Malicious insider; Machine learning; Performance evaluation; Sampling techniques
21	Budžys (2024)	Critical infrastructure ; Deep learning ;Keystroke dynamics ; Cybersecurity ; Behavioral biometrics ; Siamese neural network
22	Cai (2024)	Insider threat detection, activity-level detection, real-time detection, graph structure learning, class imbalance
23	Ding (2024)	insider threat; behavior analysis; anomaly detection; session graph; graph neural network
24	Dong (2025)	Context fusion, curriculum learning (CL), denoising diffusion probabilistic models (DDPM), insider threat detection
25	Dosh (2021)	Insider threat, CERT dataset, Cybersecurity, Attacker's psychology, IT sabotage
26	Duan (2024)	Insider threat detection ; Infrastructure security; Contextual feature ; Image representation; Multi-source feature fusion
27	Erola (2022)	Insider threat; Anomaly detection ; Cyber security
28	Fei (2023)	Insider threat detection ; Log analysis; Anomaly detection; Cyber security; User behavior analysis
29	Feng (2025)	insider threat detection; UEBA; anomaly detection; random forest
30	Ferraro (2025)	nsider threat detection ; Generative Agent-Based Modeling; Large Language Models ; Cybersecurity ; Multi-agent systems
31	Gayathri (2024) SPCAGAN	class imbalance, insider threat, hybrid deep learning models, generative adversarial networks, adversarial training, adversarial robustness
32	Gayathri R (2025) Adversarial Training	adversarial training; backdoor attacks; data poisoning; insider threat; generative models; explainable AI
33	Gayathri B (2025) ITD-GMJN	Cloud Computing, MICE, Behavioral Log Files, Golf Optimization Algorithm, LS-AE and Jordan Neural Network
34	Gonzales (2025)	User behavior analytics, context awareness, local context weighting, contiguous pattern mining, anomaly detection, Kullback Leibler divergence, n-grams, probability distribution, maximum patterns, performance metrics.
35	Greitzer (2021)	Insider threat, SOFIT ontology, expert judgments, Bayesian models, threat assessment models
36	Rahman (2022)	Unintentional insiders, threats, network, attacks, data breaches.

37	Haq (2022)	Natural language processing; insider threats; lstm; word2vec; global vectors for word representation
38	He Daojing (2022)	
39	He Daojing (2024)	Phishing attack, insider threat, deep learning, double-layer detection, simulation verification
40	Hong (2023)	Sequential activity; Graph neural networks; Insider threat; LSTM auto-encoder
41	Huang (2025)	Deep learning, insider detection, intrusion detection, masquerader attacks.
42	Hurst (2022)	Electronic health record; Machine learning; Anomaly detection; Security
43	Jaiswal (2024)	Imbalance data; Sampling; Feature engineering; Machine learning; Insider threat detection
44	Janjua (2021)	Malicious emails; Insider threat; Machine learning; Enron dataset ; TWOS dataset ; Text classification
45	Jyosthna (2021)	Organizational roles; insider threats; outlier detection; deviation score
46	Kabir (2023)	cybersecurity; insider threat; malware detection; machine learning
47	Kamatchi (2025)	Federated learning ; Rivest–Shamir–Adleman ; Elliptic curve digital signature algorithm ; Ordering points to identify the clustering structure (OPTICS) ; Federated automatic weight optimization ; Hash-based message authentication code ; Secure hash algorithm ; Insider threat detection
48	Kan (2023)	Insider threat; Behavior modeling; Particle Swarm Optimization algorithm; eXtreme Gradient Boosting; Management strategy.
49	Kim (2023)	Anomaly detection; cybersecurity; discrete wavelet transformation; insider threat classification
50	Kong (2025)	Adaptive embedding, behavior analysis, dual perspective, GloVe, insider threat, Internet of Things (IoT), sequence tagging.
51	Kotb (2025)	
52	Lavanya (2024)	Data augmentation, deep neural network, insider attack, insider threat detection, outlier estimator K-means clustering
53	Le Zinc (2021) , Exploring anomalous behavio	insider threat, anomaly detection, machine learning
54	Le and Zinc-Heywood (2021) Unsupervised ensembles	Insider threat detection, anomaly detection, ensemble learning, unsupervised learning, temporal data, dependable and robust learning.
55	Li Dongyang (2021) Image-Based	

56	Li Ximing (2023) Dual-Domain Graph	Insider threat detection, anomaly detection, graph convolutional network
57	Li (2024)	Insider threat, graph neural network, meta learning
58	Lilhore et al. (2025)	Cloud computing, Zero-Trust architecture, SmartTrust framework, LSTM deep learning, Cloud security, Reinforcement learning, Anomaly detection
59	Liu et al. (2025)	Anomaly detection; Insider threat detection; Semantic similarity; Contrastive learning; Knowledge graph
60	Manoharan et al. (2024)	Cybersecurity · Insider threat · Imbalanced dataset · Supervised learning
61	Matos et al. (2025)	Anomaly detection, Healthcare access logs, Meta-classifier, Hybrid model
62	Medvedev et al. (2025)	Cybersecurity; Keystroke dynamics; User authentication; Deep learning; Intrusion detection; Data fusion
63	Mehmood et al. (2023)	Privilege escalation, insider attack, machine learning, random forest, adaboost, XGBoost, LightGBM, classification.
64	Mehnaz & Bertino (2021)	Insider attacks, anomaly detection, file system access
65	Mladenovic et al. (2024)	Insider threat, Natural language processing, Hyperparameter optimization, XGBoost, AdaBoost
66	Nasir et al. (2021)	Insider threat, deep learning, machine learning, user behavior, information security.
67	Nikiforova et al. (2024)	Anomaly detection, clustering algorithms, data mining, information system user behavior analysis, information technology security, insider threats detection.
68	Pal et al. (2023)	Insider threat; Stacked ensemble; LSTM; GRU; Attention model; Imbalanced data; Equally-weighted random sampling
69	Patel & Iyer (2025)	Deep Learning (DL); Anomaly Detection (AD); Siamese Convolution Neural Network (SCNN); Deep Neural Network (DNN); Frequent Pattern (FP) growth algorithm
70	Peccatiello et al. (2023)	Insider threat detection, data stream, machine learning, one-class classification
71	Pérez-Gomariz et al. (2025)	Anomaly detection; Advanced persistent threat ; Lateral movements; Cybersecurity; knowledge graphs ; Natural language processing ; Transformers
72	Prasad et al. (2024) imbalanced Data resolution	Insider Threat Detection, Data Imbalance, Over Sampling, Under Sampling, Machine Learning, Ensemble Learning
73	Prasad et al. (2025) Hybrid Machine Learning	Insider threat detection, Hybrid model, Generative AI (DAE, VAE), Feature fusion, CERT dataset, Resampling.
74	Prasad et al. (2025) Deep Learning Techniques	Insider threat Detection, Behavioral Analysis, Machine Learning, CERT, ANN

75	Qawasmeh & AlQahtani (2025)	insider threats; machine learning; real-time detection; user behavior classification; risk assessment; anomaly detection; dynamic profiling
76	Qi et al. (2025)	Insider threat detection ; Temporal heterogeneous graph ; Graph neural network ; Attention neural network
77	Racherache et al. (2023)	Cyber-persona; user profiling; Insider threats detection ; Anomaly detection; Deep learning ; Machine learning
78	Randive et al. (2023 a) pattern-based approach	Insider threat; Multi-perspective feature selection ; Class imbalance ; Image-based feature representation ; Wavelet CNN; Insider threat classification
79	Randive (2023b) MWCapsNet	Cybersecurity ; Insider threat ;Image representations; Multi-level wavelet decomposition; Capsule network ; Classification
80	Rauf et al. (2021)	Insider threats ; Cyber threat intelligence ; Threat analytics ; Behavioral analysis ; Bio-inspired design; Policy regulation ; Access control ; Formal verification
81	Saminathan et al. (2023)	insider; threat; detection; autoencoder; artificial neural network; cyber security
82	Bin Sarhan & Altwaijry (2023)	insider threat; deep learning; anomaly detection
83	Senevirathna et al. (2025)	Deep learning; physical security; human behavior analysis; security operation centers; threat profiling
84	Singh et al. (2022) Multi Fuzzy Classifier	User Behavior Analysis ; Insider Threat Detection ; Isometric Feature Mapping ; Emperor Penguin ; Algorithm ; Term Frequency-Inverse Document Frequency ; Multi Fuzzy Classifier
85	Singh et al. (2023) Hybrid Learning Approach	Insider-Threat-Detection ; User behaviour analysis ; Pattern matching ; Bi-directional long short term memory ; Artificial neural network ; Support vector machine ; Anomaly detection
86	Sivakrishna et al. (2025)	Bayesian optimization ; cybersecurity ; data imbalance ; feature engineering ; insider threat ; XGBoost
87	Song et al. (2024)	Insider threat detection; Behavior pattern mining; Time information, User adaptive; Deep learning
88	Tabassum et al. (2024)	Healthcare, Anomaly detection, Insider threats, Electronic Health Records(EHRs), Machine learning
89	Tao et al. (2023) Temporal convolutional network	Insider users ; Mouse dynamics ; Feature extraction ; Temporal convolutional network ; Efficient channel attention mechanism ; Authentication
90	Tao et al. (2025) Test-Time Teaining	Insider threats ; Test-Time training ; Residual network ; Efficient channel attention mechanism
91	Tian Zhihong et al. (2024) Dempster-Shafer Theory	Deep learning ; Insider threat ; Network security ; Recurrent neural networks
92	Tian Tian et al. (2025) specific threat	Insider threat, Specific threat scenarios, Multi-attention mechanism, Behavior sequences, User behavior patterns

	scenarios	
93	Villarreal-Vasquez et al. (2023)	Anomaly detection, endpoint detection and response (EDR), high-dimensional data, insider threats, long short-term memory (LSTM), order-aware recognition (OAR) problem, sequence analysis, variable-length system activity event sequences
94	Wall & Agrafiotis (2021)	Insider threat detection, Bayesian networks, behavioral analysis
95	Wang Cheng & Zhu (2022)	Anomaly detection, behavior modeling, data enhancement, network representation learning
96	Wang & El Saddik (2023) DTITD	Digital twin, cybersecurity, insider threat, deep learning, transformer, BERT, RoBERTa, GPT-2, data augmentation, artificial intelligence, machine learning, UEBA.
97	Wang Jiarong, et al. (2023) Deep Clustering	insider threat; multi-source user behaviors; deep clustering
98	Wang et al. (2022) Fusion Model	Internet of things; insider threat; web browsing behavior; anomaly detection
99	Wang et al. (2024) FeD ITD	Cybersecurity, insider threat, deep learning, transformer, BERT, RoBERTa, XLNet, DistilBERT, GPT, data augmentation, artificial intelligence, machine learning, pre-trained LLM, PETuning, adapter, LoRA, BitFit, LLM, NLP.
100	Wei Yichen et al. (2021) Unsupervised Learning	Insider threat prediction ; Proactive forensic investigation ; Unsupervised deep learning ; Autoencoder ; Hypergraph
101	Wei Zhiyuan et al. (2024) E-Watcher	Behavioral analysis ; Information gain ; Insider threat detection ; Machine learning ; Hybrid detection
102	Wen et al. (2023)	Internal threat ; Sentiment network analysis ; SVD ; Eigenvector centrality
103	Xiao Junchao et al. (2023) Graph Neural Network	Anomaly detection, insider threat, graph neural network
104	Xiao Haitao et al. (2024) Unveiling shadows	Insider threat detection; Statistical analysis; Sequential analysis; Deep learning
105	Xiao Fengrui et al. (2025) SENTINEL	Insider threat detection, graph neural network, anomaly detection, behavior interaction graph (BIG)
106	Ye Xiaoyun et al. (2025) Federated Learning	Insider threat detection; Federated learning; Numerical data visualization; Behavior log analysis
107	Yi & Tian (2024)	representation learning; unsupervised ensemble; outlier scoring; principal component analysis; extreme gradient boosting
108	Yumlembam et al. (2025)	Anomaly Detection, Attention Mechanism, BiLSTM Cybersecurity, Deep Learning, Graph Convolutional Network (GCN), Insider Threat Detection Machine Learning, Temporal Analysis, User Behavior Analysis

109	Zhang et al. (2021)	
110	Zhu et al. (2024)	Adversarial autoencoder, insider threat detection, time series data, unsupervised learning